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1 Titration Curve of a weak acid

EXPERIMENT 1

Determination of K_a for a Weak Acid

Goal

The goal of this experiment is to measure the acidity constant of a weak acid through titration with a PH meter. The overall goal is also to use the graphical method to obtain the endpoint of a titration.

Materials

- ☐ a buret
- ☐ a hot plate with magnetic stirring
- ☐ magnetic stirring rod
- ☐ two 150 mL beakers
- ☐ a PH meter
- ☐ a NaOH unknown solution
- ☐ KHP

Background

Titrations were used in previous experiments to analyze the concentration of a solution, the so-called *analyte*. Using the *titrant*, a reactant of known concentration, it is possible to determine the concentration of the analyte if the volumes are measured carefully. The titrant is added from a buret until the solution changes color due to the presence of an indicator. The indicator denotes the moment when the *equivalence point* is achieved. The equivalence point is the moment at which equivalent molar amounts of both reactants have been added.

In an acid-base reaction, the acid reacts with the base, forming salt and water. This process is called *neutralization*; the protons released by the acid are neutralized by the hydroxide ions formed in the presence of the base. For strong acids and strong bases that dissociate completely in water, the equivalence point occurs at pH=7 and, therefore, an indicator with a color transition at pH=7 is used.

However, not all acids have the same strength. Weak acids do not release as many protons as strong acids do. The strength of a weak acid, and the consequent number of protons released, depend on its dissociation constant, K_a. Because of the partial dissociation of these species, the equivalence point in a weak acid titration does not occur at pH=7. The moment at which equivalent amounts of a weak acid and a strong base (titrant) have been added will take place at pH>7 (due to the non-complete dissociation). Determining the equivalence point is necessary to know the correct indicator to be used for titrations of weak acids.

The titration curve

During the neutralization process of an acid, the pH of the solution starts at a low value - lower than 7 since it is an acid - and increases with the addition of the base. If one plots the pH versus the volume of base added, the result is not a straight line, but an S-shape curve. This shape shows that the pH changes slowly at the beginning and the end of the process, but there

is a region in the graph where the pH increases rapidly with small additions of the base. When that abrupt pH change takes place depends on the acid and the base used.

For the titration of a weak acid (analyte) using a strong base (titrant), the endpoint will be achieved at a pH higher than 7. For the titration of a weak base (analyte) using a strong acid (titrant), the endpoint will be achieved at a pH lower than 7. The titration curve for every base or acid is unique and it is useful for determining the dissociation constant (K_a or K_b) of the analyte.

The acid dissociation constant

The acid dissociation constant for a weak acid is a ratio of the amounts of products and reactants at the equilibrium position. The ratio is constant for a fixed temperature. The concentration of H^+ will always be among the products for the dissociation of an acid. Recalling that the concentration of H^+ defines the pH, the latter can be solved as follows:

$$\mathrm{HA(aq)} \quad \rightleftharpoons \quad \mathrm{H^+(aq) + A^-(aq)}$$

$$K_a = \frac{[\mathbf{H}^+][A^-]}{[HA]} \qquad [\mathbf{H}^+] = K_a \frac{[HA]}{[A^-]} \quad ; \quad pH = pK_a + log\frac{[A^-]}{[HA]}$$

This expression is known as the *Henderson-Hasselbalch equation*, where [HA] is the concentration of the weak acid, and [A^-] is the concentration of its conjugated base. It is possible to visualize the pK_a from the titration curve by using the Henderson-Hasselbalch equation.

First, notice that at the equivalence point, the same amounts of acid and base have reacted and there will be no acid; [HA] = 0. But *halfway* to that point, when half of the acid has been converted to the same amount of its conjugated base, the concentrations of the two will be equal, [A^-]=[HA]. The fraction in the previous expression becomes equal to 1 and, since log 1 = 0, then pH = pK_a.

$$pH = pK_a + log\frac{[A^-]}{[HA]} \quad ; \quad pH = pK_a + log1 \quad ; \quad pH = pK_a$$

The equivalence point would correspond to the inflection point of the titration curve. By reading the volume of base added at that point, it is possible to calculate the volume needed to reach the halfway point ($V_{eq}/2$). The pH when that volume of base was added is taken from the the titration curve and corresponds to the weak acid's pK_a.

The location of the equivalence point can be estimated by fitting two straight lines to the straight sections of the curve, as depicted in Figure 1, and a third line equidistant to the previous two. The point at which the third line and the titration curve intersect is the equivalence point.

The experiment

The experimental part consists of the titration only. The data will be used later to plot the curve. From the plotted curve the equivalence point and the acid pK_a should be calculated.

The weak acid studied in this experiment is potassium hydrogen phthalate, $KHC_8H_4O_4$ (KHP). The base used as a titrant is sodium hydroxide (NaOH), a strong base. It is very important to obtain precise measurements of the volumes and masses used during this experiment.

The analyte (KHP) will be dissolved in a beaker. The titrant (NaOH) will be added from a buret. A pH meter will be used to monitor the pH and magnetic stirring will be used during the measurement to homogenize the solution pH.

Example

A solution of hydrogen cyanide (HCN), a weak acid, is titrated using NaOH, a strong base. The pH is monitored during the titration using a pH-meter. The following plot is obtained:

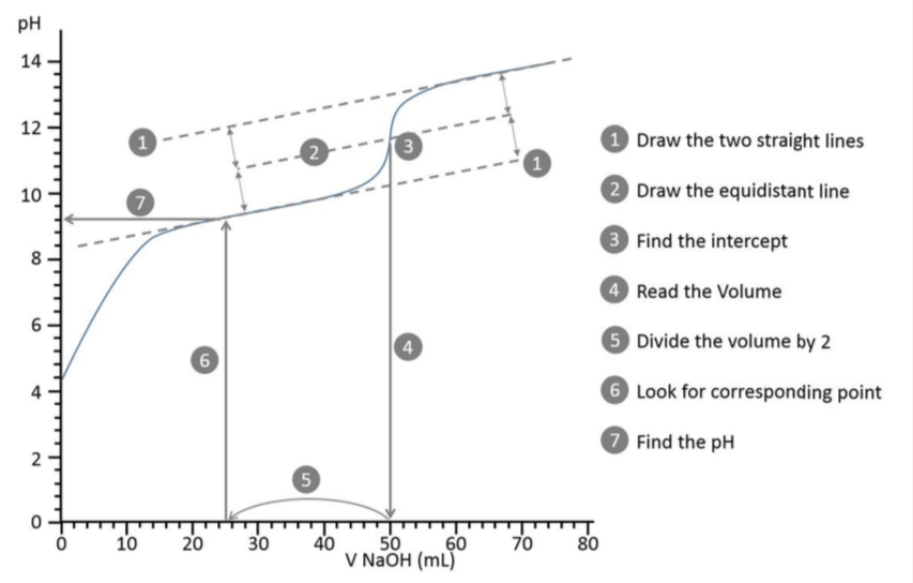

Figure 1: Titration curve of a HCN solution using NaOH. The numbers indicates the steps to follow as indicated in the answer.

Use the graphical method to identify the equivalence point for the titration, the volume of base added at the equivalence point, the midpoint, and the HCN K_a.

Answer: The equivalence point is obtained from the graph. (1) Draw the lines for the 2 straight sectors. (2) Draw the equidistant line. (3) Identify the equivalence point were the equidistant line intercepts the curve. (4) Draw a vertical line to find the volume; 50.0 mL. (5) The half-way point would be: 50.0 mL/2 = 25.0 mL. (6) Draw a vertical line up to the curve. (7) From the curve draw an horizontal line to find the pH; 9.21. The pK_a = 9.21 (7), and therefore $K_a = 10^{-9.21} = 6.2 \times 10^{-10}$

Procedure

Preparing the setup

☐ *Step 1:* – Obtain a 50 mL buret, rinse it with distilled water, and verify that it operates correctly (you should be able to add drops as well as larger volumes).

☐ *Step 2:* – Obtain a hot plate with magnetic stirring, a magnetic stirring rod, a stand, two 200 mL beakers, and a pH meter.

☐ *Step 3:* – Arrange the setup so the beaker sits on the hot plate, the pH meter is to its left with the probe inserted in one beaker, and the buret tip is also slightly inserted into the beaker. The number markings on the buret should be facing you, and the valve should be accessible and operative.

☐ *Step 4:* – Ask your instructor whether to use a NaOH unknown or the already prepared NaOH solution. Rinse the buret with a small amount of the NaOH solution. Discard the rinsing liquid. Fill the buret with fresh NaOH solution. Discard the first 3-4 mL (the volume of solution in the tip of the buret used for the rinsing).

Good Lab Practice

- ✍ Burets are read from top to bottom.
- ✍ The volume added is calculated by difference. It is not necessary to start from 0 mL.
- ✍ Use a funnel to fill the buret but remove it always before the first reading. Drops from the funnel will affect the measurements.
- ✍ Do **never** refill the buret with the Erlenmeyer/beaker of the analyte below it. A spill would ruin the titration.

Preparing the KHP solutions

☐ *Step 1:* – Put one of the beakers on the scale and set it to zero (T/O). The beaker must be dry. Remove the beaker from the scale to add a small amount of KHP. You need an amount between 0.45 and 0.55 grams. Record the value with all the decimals.

☐ *Step 2:* – Obtain 50.0 mL of distilled water in a graduated cylinder.

☐ *Step 3:* – Carefully add the 50.0 mL of distilled water from the graduated cylinder into the beaker containing the KHP.

☐ *Step 4:* – Repeat the procedure to prepare a second solution in a second beaker.

Measuring

☐ *Step 1:* – Add the magnetic stirrer to one of the beakers, put the beaker on the hotplate, and set a gentle stirring speed. Do not start the heat on the plate!

☐ *Step 2:* – Clean the pH-meter probe before immersing it in the KHP solution.

☐ *Step 3:* – Read and record the initial pH before adding any NaOH.

☐ *Step 4:* – Start adding the titrant (the base). After every addition (see next step) record the **buret reading** (not the volume added) and the corresponding pH measurement.

☐ *Step 5:* – NaOH should be added initially in 1 mL increments, but at a certain moment, additions should be reduced to 0.2 mL portions. Note that the pH will change very slowly at the beginning and the end of the titration. However, around the equivalence point, the pH will change extremely fast with small additions of the titrant.

☐ *Step 6:* – Read the buret and record the volume with all the digits given by the buret plus the estimated digit.

☐ *Step 7:* – Add the titrant in 1 mL portions, making sure the buret readings are precise.

☐ *Step 8:* – After every addition, estimate how much the pH changed from the previous pH measurement. If the pH has changed by less than 0.3 pH units, add 1 mL of titrant in the next step. If the pH has changed by 0.3 pH units or more, begin adding the titrant in 0.2 mL portions.

☐ *Step 9:* – When adding 0.2 mL portions, if the pH change starts to slow down and increases by less than 0.3 pH-units between measurements **for at least 3 measurements in a row**, then you can start adding 1 mL portions again.

☐ *Step 10:* – When the pH exceeds 11.5, perform four additional measurements in 2 mL increments, then stop the titration.

☐ *Step 11:* – Do a second titration trial. Clean and reuse the magnetic stirred; in the results section, plot by hand the titration plot.

Calculations

(1) This is the mass of your sample.

(2) These are the moles on your sample:

$$n_{\text{KHP}} = \frac{(1)}{204.22}$$

(3) These are the molarity on your acid sample:

$$c_a = \frac{(2)}{50} \times 10^3$$

(4) This is the burette reading in mL.

(5) This is the PH as given by the PH meter, and (6) is the cumulative volume of base added.

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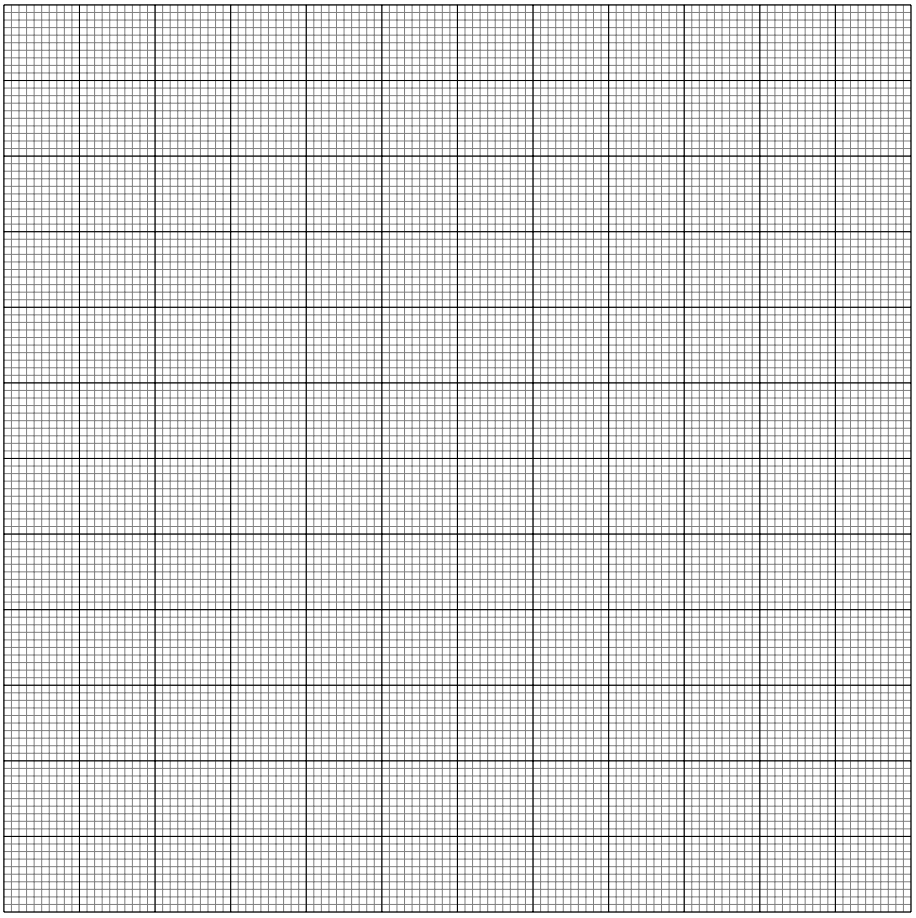
Pre-lab Questions

Determination of K_a for a Weak Acid

1. The data below represents the titration of a weak base with a strong acid. Use the data below to answer the following questions:

V_{HCl}^{Added} mL	PH	V_{HCl}^{Added} mL	PH	V_{HCl}^{Added} mL	PH	V_{HCl}^{Added} mL	PH
0	11.12	45.01	8.3	49.97	6.1	50.32	3.5
0.71	10.9	45.95	8.2	49.98	5.9	50.4	3.4
1.48	10.7	47.89	7.9	49.99	5.7	50.63	3.2
1.96	10.6	48.31	7.8	50.00	5.4	50.8	3.1
4.03	10.3	48.65	7.7	50.01	5	51.01	3
6.18	10.1	49.31	7.4	50.02	4.7	51.27	2.9
11.06	9.8	49.45	7.3	50.03	4.5	51.61	2.8
13.18	9.7	49.56	7.2	50.04	4.4	52.04	2.7
18.12	9.5	49.72	7.0	50.06	4.2	52.58	2.6
20.85	9.4	49.78	6.9	50.08	4.1	53.27	2.5
26.58	9.2	49.83	6.8	50.1	4	54.15	2.4
29.41	9.1	49.89	6.6	50.13	3.9	55.28	2.3
37.02	8.8	49.91	6.5	50.16	3.8	56.73	2.2
40.94	8.6	49.93	6.4	50.2	3.7	58.63	2.1
42.53	8.5	49.95	6.3	50.25	3.6	61.11	2

1.1. Plot PH (vertical axis) vs. volume of titrant added (horizontal axis), making sure you include all points. Identifying the initial, mid, and endpoints.



1.2. Fill the table below based on the plot above:

Initial Point		Midpoint		Endpoint	
V_{HCl}^{Added}	PH	V_{HCl}^{Added}	PH	V_{HCl}^{Added}	PH
0					

1.3. Use the midpoint to compute the basicity constant of the titrate given that at that point $PH = PK_b$. Write down the K_b value under the graph.

2. Calculate the PH of a 0.5M solution of an acid with $K_a = 1.4 \times 10^{-5}$.

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Results
EXPERIMENT 1

Determination of K_a for a Weak Acid

c_{NaOH} (M) = _____ Unknow # = _____

Trial 1			Trial 2		
KHP mass (g)	(1)	_____	KHP mass (g)		_____
moles of KHP	(2)	_____	moles of KHP		_____
c_{KHP}	(3)	_____	c_{KHP}		_____
Buret reading (4)	pH (5)	V^{Added}_{NaOH} (mL) (6)	Buret reading (4)	pH (5)	V^{Added}_{NaOH} (mL) (6)
_____	_____	0	_____	_____	0
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

Buret reading	pH	V_{NaOH}^{Added} (mL)

Buret reading	pH	V_{NaOH}^{Added} (mL)

Buret reading	pH	V_{NaOH}^{Added} (mL)

Buret reading	pH	V_{NaOH}^{Added} (mL)

Buret reading	pH	V_{NaOH}^{Added} (mL)

Buret reading	pH	V_{NaOH}^{Added} (mL)

Experimental Data						
	Initial Point		Midpoint		Endpoint	
	V_{NaOH}^{Added}	PH	V_{NaOH}^{Added}	PH	V_{NaOH}^{Added}	PH
Trial 1	0					

Graph 1 PH (Y axis) vs. V_{NaOH}^{Added} (X axis). Replicate 1

Experimental Data						
	Initial Point		Midpoint		Endpoint	
	V_{NaOH}^{Added}	PH	V_{NaOH}^{Added}	PH	V_{NaOH}^{Added}	PH
Trial 2	0					

Graph 2 PH (Y axis) vs. V_{NaOH}^{Added} (X axis). Replicate 2

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Post-lab Questions

Determination of K_a for a Weak Acid

1. Preparing a NaOH solution, starting from a solid NaOH sample, is difficult because NaOH is very hygroscopic (absorbs water from the air). This affects the mass measured and therefore the real concentration expected. The NaOH solution used was estimated to be 0.10 M but it might not be correct. Using the information from the titration (mass of the phthalate), and knowing that the acid and the base react 1:1, calculate the real concentration from each trial, and the mean value.

	Trial 1	Trial 2
KHP mass	_____	_____
KHP moles	_____	_____
V_{NaOH} @ equiv.	_____	_____
Real c_b, [NaOH]	_____	_____

Mean c_b, [NaOH] _____

2. Calculate the volume of NaOH for the halfway point (half of the volume added at the equivalence point) and locate in the graph the pH for the solution after that volume was added. Use the value of the pH to calculate the KHP pK_a.

	Trial 1	Trial 2
V_{NaOH} @ halfway	_____	_____
pH @ halfway	_____	_____
pK_a	_____	_____

Mean pK_a _____

3. Using the pK_a mean value, calculate KHP's K_a. Write the whole expression for K_a.

4. Using the KHP's K_a and its initial concentration, calculate the theoretical pH before any NaOH was added, and compare the calculated values to the pHs measured experimentally.

	Trial 1	Trial 2
KHP moles	_____	_____
c_a, [KHP]	_____	_____
Theoretical pH	_____	_____
Experimental pH	_____	_____

5. Using the K_a for KHP, calculate K_b for its conjugated base, $C_8H_4O_4{}^{2-}$, given that $K_w = K_a \times K_b$.